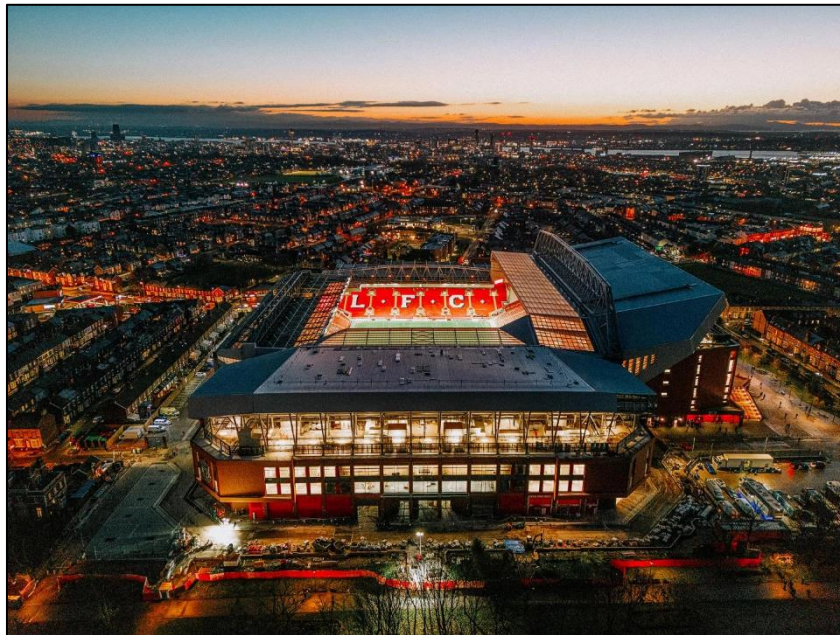


GEOS 450: Enterprise and Web GIS

Web Maps and Web Mapping Applications

Assignment 1



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Web-Based GIS Platform Comparison

1. ArcGIS Online

ArcGIS Online is Esri's cloud-based web GIS platform. It lets users create, store, analyze, and share spatial data using web maps and web applications. It is part of the larger Esri system and connects with tools like ArcGIS Pro, StoryMaps, Dashboards, Field Maps, and Experience Builder. ArcGIS Online is useful for users who need mapping, data hosting, sharing, and basic analysis in one place. Esri describes ArcGIS Online as a platform for creating interactive web maps and GIS web apps (Esri, n.d.).

1.1 Functionality

ArcGIS Online supports hosted feature layers, basemaps, pop-ups, labels, filters, bookmarks, and basic editing. Users can add common data formats such as CSV files, shapefiles, and GeoJSON files. It also includes app-building tools like StoryMaps, Dashboards, Instant Apps, and Experience Builder. Many of these tasks can be done without coding, which makes the platform easier for users who do not have web development experience.

1.2 Symbology

ArcGIS Online has built-in tools for styling points, lines, and polygons. Users can symbolize data by category, quantity, size, colour, or attribute values. Pop-ups can also be edited so users can control what information appears when someone clicks on a feature (Esri, n.d.).

1.3 Security and Sharing

ArcGIS Online uses account-based security. Content can be kept private, shared with groups, shared with an organization, or made public. Roles and privileges also control what users are allowed to create, edit, publish, or manage (Esri, n.d.).

Sharing is one of the main strengths of ArcGIS Online. Maps, layers, and apps can be shared through links, groups, or public access. One limitation is that all layers in a web app need the correct sharing settings. If one layer is private, some users may not be able to see the full map.

1.4 Analysis

ArcGIS Online includes basic spatial analysis tools. These include buffering, overlay, summarizing data, finding nearby features, and creating travel areas. It works well for common web-based GIS tasks. However, more advanced analysis, large datasets, raster work, and detailed editing are usually better done in ArcGIS Pro.

1.5 Credit Usage and Cost

ArcGIS Online usually requires an Esri account or organizational subscription. It also uses credits for activities such as storage, geocoding, routing, analysis, and other services (Esri, n.d.). This gives users cloud hosting and built-in GIS tools, but it also means ArcGIS Online is not fully free or open source.

1.6 Pros of ArcGIS Online

ArcGIS Online is easy to use and does not require coding for most basic tasks. It includes data hosting, web maps, app builders, sharing tools, and security settings in one platform. It is especially useful for organizations that already use Esri products.

1.7 Limitations of ArcGIS Online

ArcGIS Online requires an Esri account and may consume credits for storage and analysis. It is also less customizable than fully coded web mapping applications. Some advanced design or interactive functions may require other Esri products, developer tools, or custom coding.

2. Leaflet

Leaflet is an open-source JavaScript library used to create interactive web maps. It is a lightweight, mobile-friendly open-source library for interactive maps. Leaflet is not a complete GIS platform. It does not include built-in data hosting, user accounts, analysis tools, or app templates. Instead, it is designed with simplicity, performance and usability in mind for adding interactive web maps to third party applications (Leaflet, n.d)

2.1 Functionality

Leaflet can display basemaps, markers, pop-ups, lines, polygons, and GeoJSON layers. It is commonly used by developers who want more control over the design and behaviour of a web map. Leaflet can add GeoJSON objects, a format for representing simple geographical features with their non-spatial attributes, directly to maps and styled using JavaScript (Leaflet, n.d)

2.2 Symbology

Leaflet supports custom styling through JavaScript and CSS. This gives developers strong control over colours, markers, line widths, labels, and pop-ups. However, unlike ArcGIS Online, Leaflet does not provide a full graphical interface for styling data. The user must usually write code or use plugins (GeeksforGeeks, 2024).

2.3 Security and Sharing

Leaflet itself does not provide built-in user accounts, group sharing, or organization-level security. Because the web map is hosted using a third-party service, security is outsourced to that service and how the data services are managed. This gives flexibility, but it also means the developer is responsible for hosting, permissions, and data access.

2.4 Analysis

Leaflet is mainly a display and interaction library. It is not designed as a full GIS analysis platform. Spatial analysis usually needs to be completed before publishing the data, or added through other libraries and plugins.

2.5 Credit Usage and Cost

Leaflet itself is free and open source. There are no Leaflet credits or licence fees. However, there may still be costs for web hosting, domain names, basemap services or cloud storage

2.6 Pros of Leaflet

Leaflet is free, lightweight, flexible, and highly customizable. It is a strong option for users with web development skills who want full control over the appearance and behaviour of a web map. Its functionality can also be expanded using plugins and because it is open source, there are a lot available on places like GitHub (GeeksforGeeks, 2024).

2.7 Limitations of Leaflet

Leaflet requires coding knowledge and does not include built-in GIS hosting, organization sharing, web analysis tools, or web application templates. It is best used as a tool to effectively display highly customized maps on other platforms.

Summary

ArcGIS Online and Leaflet are both used for web mapping, but they are made for different purposes. ArcGIS Online is a complete cloud-based GIS platform, while Leaflet is an open-source JavaScript library for building custom web maps. AGOL is more powerful and functional, while Leaflet is highly customizable and portable.

Web Map Creation

3. Web Map Purpose

The web maps I developed support the StoryMap by organizing the project into clear spatial themes. Each map focuses on one part of Liverpool FC's geography, such as local cultural context, domestic club locations, league title success, European final victories, first-team player countries, and the birthplaces of selected legends. Together, these maps help the StoryMap move from the city of Liverpool to national, European, and global scales.

Multiple web maps were created to keep the StoryMap clear and easier to follow. Each map focuses on one main theme, which avoids putting too many layers on one map, which could make it crowded and

confusing. Using separate maps also allows each dataset to have the right scale, symbols, and pop-ups. This makes each section of the StoryMap more focused and organized.

3.1 Data Used in the Web Map

Web Map	Hosted Feature Service	Geometry	Display Method	Purpose
<i>English Premier League Stadiums</i>	Premier_League_Clubs	Points	Stadium locations shown as point symbols	Shows where Premier League clubs are located across England.
<i>Total Number of English League Titles</i>	Premier_League_Clubs	Points	Graduated symbols based on total league titles	Compares domestic league success between English clubs.
<i>Liverpool Legends Birthplace</i>	LFC_legends_birthplaces	Points	Birthplace locations shown as point symbols	Shows the geographic origins of selected Liverpool FC legends.
<i>Liverpool FC Champions League Wins</i>	LFC_European_Wins	Points	European final venues shown with trophy-style symbols	Shows where Liverpool won its European Cup and Champions League titles.
<i>Liverpool First Team Countries</i>	LFC_Player_Countries	Polygons	Country polygons symbolized by player representation	Shows the countries represented by Liverpool's current first-team players.
<i>No separate web map created</i>	LFC_Cultural_Context	Points	Local landmarks shown as point symbols	Supports the local city context section of the StoryMap.

4. Web Map Application Purpose

Project Title: All Roads Lead to Anfield: *Mapping Liverpool FC's City, Rivals, Legends, and European Glory*

StoryMap URL: [All Roads Lead to Anfield](#)

The purpose of this StoryMap application is to show Liverpool FC's geography through a guided and interactive format. Instead of presenting each web map on its own, the StoryMap combines the datasets into one organized application. It starts with Liverpool's local cultural context and then moves into the club's domestic football landscape, league success, European victories, current first-team countries, and the birthplaces of selected legends. This format helps users see how Liverpool FC is connected to the city of Liverpool, English football, European competition, and a wider global football identity.

4.1 Home of the Reds

This section introduces the local context of Liverpool FC using the *LFC_Cultural_Context* hosted feature layer. This point-based dataset includes important football and cultural locations in Liverpool. The section uses an immersive map tour, where each stop focuses on one location and gives a short description. This helps introduce Anfield and other places that connect the club to the wider city.

4.2 The Premier League Geography

Developed to place Liverpool FC within the domestic football landscape, this section uses the English Premier League Stadiums web map and the Total Number of English League Titles web map. Both maps were created from the *Premier_League_Clubs* hosted feature layer. Infographics are used to show distance and away travel context. A swipe map compares stadium locations with league title success, so users can see both club geography and historical achievement. Users can also click the map icons to view more information about each club.

4.3 European Glory

Here we focus on Liverpool FC's European Cup and Champions League victories. This section uses the Liverpool FC Champions League Wins web map, which was created from the *LFC_European_Wins* hosted feature layer. A sidecar format is used with bookmarks, so the map zooms to each final location as the story moves forward. Embedded videos are also included to make the section more interactive and add historical context.

4.4 Liverpool Legends

This section highlights both past and present player geography. An image gallery carousel shows Liverpool legends from different eras. The section also uses a sidecar with action buttons, where each button zooms to the birthplace of a selected legend. This uses the Liverpool Legends Birthplace web map and the *LFC_legends_birthplaces* hosted feature layer. Another sidecar slide shows the Liverpool First Team Countries web map, created from the *LFC_Player_Countries* polygon feature layer. Users can click highlighted countries to see which current players come from each country, along with related player images.

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